

AI ASSURANCE FORUM 2025

ASSURANCE FOR RAPID AI ADOPTION

REPORT PARTNER

TEMASEK ×  resaro

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2025 TEMASEK × RESARO AI ASSURANCE FORUM

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FIG 1.1. 'GOLDEN HOUR' – PARTICIPANTS IN ACTIVE DISCUSSION

On 26 May 2025, the inaugural Temasek x Resaro AI Assurance Forum convened thought leaders across the AI ecosystem for a series of discussions around the theme of “Assurance for Rapid AI Adoption”. Organised by Temasek and Resaro, 35 global leaders from across industry, government, academia, and civil society gathered from the US, EMEA, and APAC. Held in Singapore, the Forum was part of the broader AI Assurance and Testing series under ATxSummit, Asia’s flagship technology event.

The Forum explored how this ecosystem engages with key dimensions of AI assurance. The morning featured panel discussions with experts across sectors:

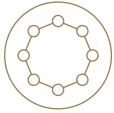
PANEL 1: OPTIMISING BUSINESS VALUE OF AI ASSURANCE

PANEL 2: OVERCOMING BARRIERS IMPEDING AI ASSURANCE AND TESTING

PANEL 3: BUILDING PRACTICAL AI ASSURANCE FRAMEWORKS

The participants contributed a selection of topics that arose out of the panel sessions, and voted on the top four that warranted greatest interest and closer examination. These four topics formed the basis of facilitated conversations in working groups after lunch.

In the afternoon, participants joined working group sessions designed as open, collaborative conversations. Each working group focused on one question, selected through a participant vote. Through active discourse, each group produced a thought piece in response to the following prompts:



COALITIONS

What should technology owners in organisations do to build internal coalitions around successful AI system deployment (including assurance) across business owners, technology owners, and compliance owners?



INVESTMENT

What tradeoffs should technology owners and business owners take into account when deciding to invest in AI assurance (or not)? Which of these tradeoffs should technology owners and business owners prioritise more when considering the context and use of AI assurance in their organisations?



DESIGN PRINCIPLES

What design principles should be implemented when building an ecosystem of AI assurance regulations, frameworks, standards, and tools for providers/deployers/testers to ensure that the end result is high-performing, high-confidence AI?



STANDARDS

Standards tend to be conceptual, general, and static – they are thus in tension with AI systems, which are necessarily concrete, context-specific, and dynamic. What measurement tools and techniques can policymakers, standards setters, and technology owners apply to develop standards that explicitly manage this tension?

This Report presents the discussions held throughout the panels and working group sessions at the Forum in three sections:

1. **Investing in AI Assurance** – Addressing how the costs and benefits of AI Assurance are perceived, how the decision to invest in AI Assurance interacts with AI adoption, and recommending an iterative and evolving mindset towards AI Assurance;
2. **Challenges Implementing AI Assurance within an Organisation** – Factors internal to an organisation that impact the adoption of AI assurance practices; and
3. **External Ecosystem Factors Influencing AI Assurance** – External influences, including policy landscapes, public expectations, and evolving regulatory frameworks, that impact the provision of AI assurance services.

With practical takeaways from their discussions presented at the end.

All discussions were held under Chatham House rules.



BRIDGING THE GAP: FROM TASK-SPECIFIC METRICS TO COMPREHENSIVE AI TESTING STANDARDS



ADAM LEON SMITH, CHAIR OF AIQI CONSORTIUM

Often, when talking about international standards for testing (or evaluation) of AI, I hear something important: “You can’t specify testing without knowing the use case”. This resonates with me, because testing is context-dependent. This is well understood conventionally. You can’t specify how to test software without knowing what it is required to do.

But in AI, that isn’t entirely true. A crucial element often goes unaddressed in these discussions: the fundamental importance of **task-specific evaluation** approaches.

THE FOUNDATION: TASK-SPECIFIC EVALUATION

An AI task represents a specific algorithmic problem – such as machine translation, classification, regression, ranking, clustering, object detection, automatic summarisation, speech recognition, pose estimation, and image segmentation. There are a lot of such tasks, but right now, experts can name about a hundred of them. These tasks are statistical in nature, yet also technology-agnostic. It doesn’t matter which technology you use for machine translation – the task retains the same objective and characteristics.

Sometimes even the actual metrics remain the same, though the requirements for how you apply them may differ.

Each task has a clear objective, with associated evaluation metrics and techniques. If a computer vision expert performs an evaluation of the correctness of an object detection algorithm for detecting rabbits, they would use exactly the same evaluation techniques if they were detecting houses. Compared to applications or use cases, the list of tasks that AI can perform is a relatively well-bounded problem space.

Of course, tasks can be joined together, used in pipelines and interact. But they represent a level of decomposition that can be considered equivalent to “component testing”. Tasks can co-exist within the same AI system or feed into one another. Each contributes to the overall performance of the AI system in the context of its intended use.

A large part (but not all) of AI assurance seeks to ensure systems are correct, robust, and free from unwanted bias. While other qualities like controllability matter, these three characteristics are both critical and statistically measurable. However, their evaluation cannot be divorced from the correctness of the specific tasks AI systems are designed to perform.

Bias and robustness measurements fundamentally derive from task-specific correctness metrics. Rather than being separate concerns, they largely represent differential assessments of how systems perform across various contexts:

Bias evaluation examines performance variations across demographic groups, sensitive attributes, or other data characteristics.

Robustness assessment analyses performance under different perturbations, distributional shifts, or unexpected conditions.

There are, of course, other ways to evaluate unwanted bias and robustness, but this is the primary approach.

WHY GENERIC APPROACHES FALL SHORT

Process-level standards provide valuable development frameworks, but they cannot effectively measure correctness, unwanted bias, or robustness without task-specific definitions. These measurements become abstract and uninterpretable without concrete, task-anchored metrics. Management system standards may specify that metrics should be defined, and testing standards may outline metric selection processes. But without task-level specificity, translating these into actionable steps remains a challenge.

The standards landscape is evolving toward more granular, task-specific approaches. ISO/IEC 4213, which provides evaluation metrics for classification tasks, is being updated to include regression, recommendation, and clustering tasks. Future standards like ISO/IEC 23282 for natural language processing and CEN-CENELEC's JT021025 for computer vision technologies will further expand this task-specific foundation. As these task evaluation approaches become standardised and more complete, they can be paired with benchmarks to provide a rich, comparative assessment.

THE NEXT CRITICAL STEP: ISO/IEC 42119-8

Building directly on these task-level foundations, the newly approved ISO/IEC 42119-8 standard represents a significant advancement in AI testing methodology. This technical specification, *"Quality assessment of prompt-based text-to-text systems that utilise generative AI,"* addresses a critical gap in current AI assurance practices. It specifically builds on ISO/IEC 23282, which contains the evaluation requirements and metrics for natural language processing, which of course includes text processing.

The new standard specifically targets prompt-based text-to-text AI systems through an adversarial, dynamic process using human and model ingenuity to probe systems for vulnerabilities, biases, and misuse potential through prompt-based attacks. While emphasising red teaming, the standard recognises and integrates other assessment approaches like benchmarking, drawing connections between various quality assessment methods and building on existing SC 42 content. The standard applies task-level evaluation principles specifically to text-to-text systems, addressing the unique challenge that generative AI outputs are less bounded and predictable than those of traditional AI systems.

This development shows how task-specific evaluation principles can be extended to address emerging AI technologies. By building on the foundational understanding that bias and robustness metrics derive from task-specific correctness measures, ISO/IEC 42119-8 will provide concrete, implementable guidance for one of today's most challenging AI assurance scenarios. Testing, bias, and robustness standards must continue to build upon task-specific functional correctness standards rather than existing in isolation. ISO/IEC 42119-8 aims to demonstrate this principle in action.

THE PATH FORWARD

The evolution of international standards toward task-specific AI assurance represents more than an incremental improvement – it creates a coherent framework where quality dimensions are understood as interconnected aspects of system performance, enabling more effective development, deployment, and governance of AI technologies across their full spectrum of applications. It is important to understand that this does not replace the need for vertical- or application-specific metrics, evaluation, or standards. When you put tasks together, things can change, and more work needs to be done to understand the relationship between AI systems, their use cases, and the tasks they are composed of.